



Empirical distribution of daily stock returns of selected developing and emerging markets with application to financial risk management

Juraj Pekár¹ · Mário Pčolár¹

Accepted: 9 August 2021 / Published online: 27 September 2021

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2021

Abstract

The paper deals with the analysis of the empirical distribution of returns of selected market indices of developed and developing markets in relation to applications in the field of financial risks. The probability distribution of returns plays a key role for both the theoretical and practical part of financial theory. The assumption of a normal distribution of returns, which is crucial for several financial theories, is rejected in line with our analysis of the daily returns of 30 markets. From the perspective of financial risk assessment, the choice of an appropriate distribution is key to the accuracy of the achieved results. In this paper, we examine the possibility of applying alternative distributions, in order to better capture the observed leptokurtic behavior of the empirical distribution of daily returns as compared to the assumption of normality. We perform the analysis on the daily returns of stock market indices in order to examine the possible differences between the empirical distribution of developed and emerging markets. We investigate the suitability and relative performance of these alternative distributions: the generalized skewed t distribution (sgt), the generalized lambda distribution (gld), the exponential power distribution (GED), the Hansen's skewed t distribution and the Laplace distribution. The best relative performance of fit for market index returns providing the sgt_2 and gld distributions, with gld appearing to be a more prominent adept than with the perspective of relative stability of quality of fit and also quality of risk estimation.

Keywords Daily returns · Generalized lambda distribution · CVaR · Skewed generalized t-distribution · Fitting distribution

✉ Juraj Pekár
juraj.pekar@euba.sk

Mário Pčolár
mario.pcolar@euba.sk

¹ Department of Operations Research and Econometrics, University of Economics in Bratislava, Dolnozemska cesta 1, Bratislava 852 35, Slovakia

1 Introduction

One of the important parts of financial modeling as well as in estimating financial risks is the analysis of the probability distribution of returns (Krämer 2021; Massing 2019). The aim of the article is to analyze 30 stock indices from developed and emerging markets. The primary goal of the analysis is to determine the distribution of individual indices and then compare the behavior of developed and emerging markets. In financial analyzes, time series of returns and their probability distributions are used to determine the risk of assets, while the Conditional Value at Risk (CVaR) is currently often used. Another objective of the article, is the suitability of using estimated distributions to determine CVaR risk, which follows from the above. Behavior of market risk in developing and developed markets was also compared.

An important assumption in the beginnings of financial modeling was the assumption of a normal distribution of returns on which is build a large amount of financial theory. Several contributions point to the findings that the assumption of a normal distribution cannot be unambiguously rejected in the case of long horizon returns as monthly or weekly. In their article, (Glawischnig and Seidl 2013) show on monthly data that they cannot reject the hypothesis of normality in any of the tested indices. They also point out that serial correlation plays a more significant role than skewness and fat tails in monthly data. On the other hand, several contributions (some listed below) also agree that daily returns are far from the normality. The empirical distribution of daily returns on financial assets is somewhat more pointed and thicker at the tails compared to the normal distribution. Peters (1991) points out that if information does not flow into markets linearly, but rather in irregular outbursts, investors will be forced to react in the same way, which ultimately leads to a leptokurtic distribution of returns. A large number of authors, especially in the 70 s to 90 s of the last century, dealt with the search for alternative probability distributions that would more appropriately describe the leptokurtic properties of stock returns.

Many authors have addressed this question as to which distribution describes the more appropriate empirical characteristics of stock returns compared to the normal distribution. Some of the contributions are based on theoretical models of investor behavior, others adapt commonly used distributions in order to obtain the best possible power of fit. The wide range of distributions tested so far includes most of the commonly used distributions in finance, we point to some of them. Stable Paretian distribution, the application of which has been dealt with by several authors. Best known are the contributions of Mandelbrot (1963) and Fama (1965), which were among the first to point out that empirical data on price changes are not normally distributed. In both cases, they designed and tested an alternative distribution, namely the stable Paretian distribution. Praetz (1972) proposed Student-t distribution due to its better properties compared to stable Paretian distribution, namely finite Variance and known density function. On weekly data, he showed that the Students-t distribution describes the data better than the stable Paretian distribution.

In the 1990s, another approach emerges to capture the non-normal characteristics of asset returns. Using conditional Variance models such as autoregressive conditional heteroscedasticity (ARCH) type of models or the generalized autoregressive conditional heteroscedasticity (GARCH). Pioneers in this field have been e.g., Engle (1982)

and Bollerslav (1986). Another approach is to use stochastic volatility processes such as in Hull and White (1987). In our paper, we focus on the unconditional point of view.

Another of the many contributions that provided evidence that asset returns do not follow the normal law was Gray and French (1990), also confirm that empirical data have longer tails and are peakier in the middle of the distribution. They point to the findings that both t-scaled, logistic and exponential power distribution describe the characteristics of daily returns better than the normal distribution, superior fit for daily returns providing exponential power distribution. Two families of very flexible shapes were proposed by Harris and Küçüközmen (2001) namely exponential generalized beta and skewed generalized-t families of distributions. Both families provide better relative performance compared to the normal distribution. The skewed generalized-t distribution family provides better alternative from a given pair.

Among the more recent contributions, we can mention Behr and Pötter (2009), who compared the mixture of Gaussians, the generalized logF distribution and generalized hyperbolic distribution on daily and monthly data of the S&P 500 index, noting that generalized hyperbolic distribution represents the best alternative for modeling daily data, on the other hand, it is the worst alternative for monthly data, where the best model is the two component Gaussian mixture. Corlu et al. (2016) tested the generalized lambda distribution, the Johnson translation system, the normal inverse Gaussian distribution, the g-and-h distribution and the skewed students-t distribution as alternative distributions for modeling daily returns. At the daily market's indices data, they show that the distribution of daily returns best describes the generalized lambda distribution. They also point to its relative stability in terms of fitting to different shapes of distributions of daily data.

The paper is divided into six sections. In the introductory section, we briefly describe the issue of the probability distribution of daily returns on financial assets and providing a brief overview of the literature dealing with the analysis of this issue in connection with the topic of our paper. In the following section, we will present selected probability distributions that we consider as alternatives which could better describe daily returns compared to the normal distribution. In the "Data description and analysis" section, we describe in more detail the data used in the analysis itself, we present several tables and figures to make it easier to interpret selected descriptive statistics of individual data. In the next section "Goodness of fit analysis", we briefly describe the graphical analysis of goodness of fit analysis and analysis based on quantified goodness of fit statistics and information criteria. In the section "CVaR estimation analysis" we examined the suitability of estimated distributions for CVaR estimation. Subsequently, we present and interpret the achieved results in a suitable form. Finally, we present a short discussion and overview of the achieved results.

Our paper contributes to the literature by analysis of the relative quality of fit of selected probability distributions, using daily data of market indices of emerging and developed markets from the point of view of differently long subperiods. The gld and sgt_2 provide superior relative quality of fit but gld is more stable and provides, on average, better estimation quality for CVaR. We didn't find significant differences in terms of quality of fit of individual distributions between emerging and developed markets. We also found out that we can observe convergence between emerging and developed markets from the perspective of market risk.

2 Selected alternative distributions

In this chapter we will briefly present the individual selected probability distributions whose relative performance will be tested in the Sect. 4.

2.1 Generalized lambda distribution

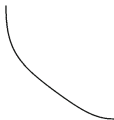
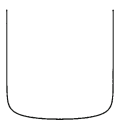
Generalized lambda distribution (gld) is a family of distributions that are characterized by a wide range of shapes that can be taken by their individual distributions. It is a modification and extension of the original Tukey's lambda distribution (Tukey 1960). In our paper, we consider FMKL parameterization (Freimer et al. 1988) defined by a quantile function with four parameters:

$$Q(y) = \lambda_1 + \frac{1}{\lambda_2} \left[\frac{y^{\lambda_3} - 1}{\lambda_3} - \frac{(1 - y)^{\lambda_4} - 1}{\lambda_4} \right] \quad (1)$$

where $0 \leq y \leq 1$. The individual parameters gld describe: λ_1 represents the position parameter, λ_2 represents the scale parameter and λ_3, λ_4 define the characteristics of the shape of tails. If $\lambda_3 = \lambda_4$ the distribution is symmetric. The individual parameters of gld are real numbers with a constraint in the case of $\lambda_2 > 0$. The k^{th} moment of FMKL gld exists only if $\min(\lambda_3, \lambda_4) > -k^{-1}$. FMKL gld parameterization allows to classify the generated distribution curves into 5 groups, described in more detail in Table 1. Dean (2013) provides a detailed overview of the issues and breakdown of the individual classes associated with generalized lambda distribution.

There are several different estimation methods for estimating FMKL gld parameters. In our paper, we consider the method of maximum likelihood estimation described in more detail in Dean (2013). In accordance with the quantified estimates of the parameters for the individual data, it follows that the adjusted FMKL gld distributions for the daily returns of the market indices of the individual markets, belong to the first class according to the classification given in Table 1.

Table 1 Characteristics of 5 classes of shapes of FKML gld parameterization

Class	I	II	III	IV	V
λ_3	$\lambda_3 < 1$	$\lambda_3 > 1$	$1 < \lambda_3 < 2$	$\lambda_3 > 2$	$\lambda_3 > 2$
λ_4	$\lambda_4 < 1$	$\lambda_4 < 1$	$1 < \lambda_4 < 2$	$1 < \lambda_4 < 2$	$\lambda_4 > 2$
Shape of curve	unimodal	monotone	U-shaped	S-shaped	unimodal
Tails	continuous		truncated		truncated
Shape					

2.2 Skewed generalized t distribution

The author who first introduced a family of continuous distributions called Skewed generalized t distribution (sgt) is Theodossiou (1998). In his article, he introduced it as a skewed version of the well-known family of generalized t distribution. Sgt is a flexible family of distributions that offers the ability to capture a wide range of different levels of skewness and kurtosis. There are currently several parameterizations of this family. In our paper we consider parameterization with 5 parameters ($\mu, \sigma, \lambda, p, q$) where sgt is defined by probability density function (pdf) in the form:

$$f(x; \mu, \sigma, \lambda, p, q) = \frac{p}{2v\sigma q^{\frac{1}{p}} B\left(\frac{1}{p}, q\right) \left(\frac{|x-\mu+m|^p}{q(v\sigma)^p (\lambda \operatorname{sign}(x-\mu+m)+1)^p} + 1 \right)^{\frac{1}{p}+q}} \quad (2)$$

where B is the beta function, μ is the location parameter, $\sigma > 0$ is the scaling parameter, $-1 < \lambda < 1$ is the skewness parameter, $p > 0$ and $q > 0$ are the parameters that control the kurtosis. Where:

$$m = \frac{2v\sigma\lambda q^{\frac{1}{p}} B\left(\frac{2}{p}, q - \frac{1}{p}\right)}{B\left(\frac{1}{p}, q\right)} \quad (3)$$

and

$$v = q^{-\frac{1}{p}} \left[\left(3\lambda^2 + 1 \right) \left(\frac{B\left(\frac{3}{p}, q - \frac{2}{p}\right)}{B\left(\frac{1}{p}, q\right)} \right) - 4\lambda^2 \left(\frac{B\left(\frac{2}{p}, q - \frac{1}{p}\right)}{B\left(\frac{1}{p}, q\right)} \right)^2 \right]^{-\frac{1}{2}} \quad (4)$$

The sgt family of continuous distributions as a special case includes a large number of commonly used distributions. A detailed overview of the individual special cases with the values of the individual parameters required to obtain the special case is provided in Fig. 1 taken from the paper (Hansen et al. 2010).



Fig. 1 Special cases of the skewed generalized t distribution

The maximum likelihood estimation method was used to estimation of the parameters of sgt distribution. In the article, we use denotation as sgt_2 for the skewed generalized t distribution.

One of the special cases of the sgt, when one of the parameters controlling the tails of the distribution is fixed to $p = 2$ is Hansen's skewed t distribution (ST). A PDF with four parameters μ, σ, λ, q has the following form:

$$f(x; \mu, \sigma, \lambda, q) = \frac{\Gamma\left(\frac{1}{2} + q\right)}{v\sigma(\pi q)^{\frac{1}{2}}\Gamma(q)\left(\frac{|x-\mu+m|^2}{q(v\sigma)^p(\lambda \operatorname{sign}(x-\mu+m)+1)^2} + 1\right)^{\frac{1}{2}+q}} \quad (5)$$

To estimation of the parameters as in the previous case, we used the method of maximum likelihood estimation, further in the paper we use denotation as sgt_1.

2.3 Exponential power distribution

Also known as generalized error distribution (GED). It is a family of continuous probability distributions which adds a shape adjusting parameter to the normal distribution. We consider the parameterization of Exponential power distribution (GED) with 3 parameters (μ, σ_p, p) , where pdf is defined in the form:

$$f(x) = \frac{1}{2\sigma_p p^{\frac{1}{p}} \Gamma\left(1 + \frac{1}{p}\right)} \exp\left(-\frac{|x - \mu|^p}{p\sigma_p^p}\right) \quad (6)$$

where $-\infty < x < \infty, -\infty < \mu < \infty$ parameter defining the location, $\sigma_p > 0$ scale parameter, $p > 0$ shape parameter which affects the width of the tails of the distribution.

A special case that is included within this family is the Laplace distribution if $p = 1$ and also the normal distribution if $p = 2$. The PDF of the Laplace distribution with two parameters (μ, σ) has the following form:

$$f(x) = \frac{e^{-\frac{|x-\mu|}{v\sigma}}}{2v\sigma} \quad (7)$$

where $v = 1/\sqrt{2}$. We use the maximum likelihood estimation method to estimation of the distribution parameters.

3 Data description and analysis

The data used in our paper represent daily data from 30 countries on the development of their blue-chip stock market indices in local currencies. Namely the following countries and market indices: Germany—DAX, United States of America—S&P 500, Sweden—OMX30, Japan—Nikkei225, Australia—S & P / ASX200,

Italy—FTSE MIB, Austria—ATX, New Zealand—NZX50, Great Britain—FTSE100, Israel—TA125, France—CAC40, Canada—S & P / TSX, Belgium—BEL20, Denmark—OMXC20, Netherland—AEX, India—BSE SENSEX, Brazil—IBOVESPA, Indonesia—IDX Composite, South Korea—KOSPI, Argentina—MERVAL, Mexico—IPC, Taiwan—TAIEX, China—SSE Composite, Croatia—CROBEX, Egypt—EGX30, Hong Kong—HSI, Chile—IPSA, Pakistan—KSE100, Russia—MOEX, Philippines—PSE.

Table 2 describes the structure of the analyzed data of different developing and developed markets. For individual markets, we present the name of the country, the usual denotation of the selected market index, the denotation that we use in our analysis and the start date of each time series, because the daily stock market indices data are not available in the same length for all countries. The end date is for all countries 31.12.2019 or 30.12.2019 depending on the country.

Daily closing positions are collected from publicly available sources at websites finance.yahoo.com and investing.com. Daily returns are quantified as continuously compounded stock market returns, i.e., as the first differences of natural logarithms of each series:

$$R(t) = \ln\left(\frac{S_t}{S_{t-1}}\right) \quad (8)$$

where S_t represents the close position of the stock index at time t and at time $t - 1$, respectively. In addition to the entire length of the time series we also consider subperiods with periodicity of 2 years, 5 years and 10 years, for each market. For example, one of the subperiods for each periodicity is: 1990–1991 for 2 years, the total number of individual series is 389; 1990–1994 for 5 years, the total number of individual series is 164; 1990–1999 for 10 years total number of individual series is 86. In the article we consider the whole data interval of individual markets as well as individual subperiods. In the following parts of the paper, we will present the achieved results for both the whole interval and partial intervals.

In Tables 3 and 4 we present relevant quantified descriptive statistics associated with the empirical distribution of daily returns of stock market indices of the analyzed developing and developed markets for whole period. In both cases, the table captures the first four moments of each series, namely the mean, standard deviation, skewness and kurtosis, variance, min, max, the p-value of Jarque–Bera normality test quantified at 1% significance level and number of observations.

Descriptive statistics in both developed and developing markets suggest a possible departure from the hypothesis of a normality of daily returns. In all cases, if we analyze the whole period, the quantified values of kurtosis are significantly above the value of 3, which is characteristic for the normal distribution. Significantly higher kurtosis suggests that empirical distributions of daily returns of individual markets will have ticker tails of the distributions compared to the normal distribution.

In the case of skewness, the characteristic skewness for the normal distribution is at level 0, the quantified skewness of each series indicates that, on average, the data are fairly symmetrical. It is also true that, on average, the daily returns of market indices are characterized by a negative skewness. Such findings can be observed in Fig. 2,

Table 2 Characteristics of time series of the selected developing and developed markets

Developed markets	Indices	Our abbre	Period (from)	Emerging markets	Indices	Our abbre	Period (from)
Netherlands	AEX	AEX	12.10.1992	India	BSE SENSEX	BSESN	01.07.1997
Austria	ATX	ATX	11.11.1992	Croatia	CROBEX	CROBEX	23.01.2001
Belgium	BEL20	BFX	09.04.1991	Egypt	EGX30	EGX30	04.01.1998
France	CAC40	FCHI	01.03.1990	Hong Kong	HSI	HSI	02.01.1990
Germany	DAX	GDAXI	02.01.1990	Brazil	IBOVESPA	BVSP	27.04.1993
Great Britain	FTSE100	FTSE	02.01.1990	Indonesia	IDX	JKSE	06.04.1990
Italy	FTSEMIB	FTSEMIB	31.12.1997	Mexico	IPC	MXX	08.11.1991
Japan	Nikkei225	N225	02.01.1990	Chile	IPSA	IPSA	27.09.1993
New Zealand	NZX50	NZX50	03.01.2001	South Korea	KOSPI	KS11	01.07.1997
Sweden	OMX30	OMXS30	02.01.1990	Pakistan	KSE100	KSE	25.05.1994
Denmark	OMXC20	OMXC20	12.02.2001	Argentina	MERVAL	MERV	08.10.1996
Australia	S&P/ASX200	AXJO	23.11.1992	Russia	MOEX10	MOEX10	20.03.2001
Canada	S&P/TSX	GSPTSE	02.01.1990	Philippines	PSE	PSEIPS	02.01.1990
USA	S&P500	GSPC	02.01.1990	China	SSE	SS	02.07.1997
Israel	TA125	TA125.TA	08.10.1992	Taiwan	TAIEX	TWII	02.07.1997

Table 3 Descriptive statistics of stock market returns of developed markets

Tickers	Mean	Std	Var	Skewness	Kurtosis	Max	Min	JarquBera	Obs
AEX	0.00023	0.01316	0.00017	-0.17630	9.70647	0.10028	-0.09590	0	6939
ATX	0.00021	0.01316	0.00017	-0.40068	10.34102	0.12021	-0.10253	0	6712
AXJO	0.00022	0.00937	0.00009	-0.47554	8.62736	0.05724	-0.08704	0	6851
BFX	0.00016	0.01122	0.00013	-0.06979	9.69348	0.09334	-0.08319	0	7278
FCHI	0.00016	0.01352	0.00018	-0.07730	7.77991	0.10595	-0.09472	0	7564
FTSE	0.00015	0.01081	0.00012	-0.12974	9.17564	0.09384	-0.09265	0	7605
FTSEMIB	-0.00004	0.01557	0.00024	-0.53301	11.17117	0.10874	-0.18546	0	5833
GDAXI	0.00026	0.01392	0.00019	-0.12774	7.76232	0.10798	-0.09628	0	7578
GSPC	0.00029	0.01098	0.00012	-0.26710	11.85507	0.10957	-0.09470	0	7557
GSPTSE	0.00019	0.00960	0.00009	-0.72200	14.41790	0.09370	-0.09788	0	7620
N225	-0.00007	0.01497	0.00022	-0.16042	8.53605	0.13235	-0.12111	0	7367
NZX50	0.00040	0.00721	0.00005	-0.49358	11.19752	0.06937	-0.06353	0	5089
OMXC20	0.00029	0.01262	0.00016	-0.32370	8.61140	0.09496	-0.11723	0	4912
OMXS30	0.00028	0.01438	0.00021	0.01869	7.53751	0.11023	-0.11173	0	7727
TA125.TA	0.00040	0.01373	0.00019	-0.46606	8.03929	0.07692	-0.10538	0	5540

Table 4 Descriptive statistics of stock market returns of developing markets

Tickers	Mean	Std	Var	Skewness	Kurtosis	Max	Min	JarquBera	Obs
BSEN	0.00041	0.01482	0.00022	-0.09615	9.86426	0.15990	-0.11809	0	5534
BVSP	0.00128	0.02219	0.00049	0.49215	13.82663	0.28833	-0.17208	0	6597
CROBEX	0.00012	0.01160	0.00014	-0.34219	23.40493	0.14779	-0.10764	0	4907
EGX30	0.00044	0.01659	0.00028	-0.37626	11.94709	0.18369	-0.17992	0	5567
HSI	0.00031	0.01556	0.00024	-0.02261	12.69856	0.17247	-0.14735	0	7402
IPSA	0.00029	0.01139	0.00013	-0.33835	17.16401	0.11803	-0.15216	0	6741
JKSE	0.00032	0.01417	0.00020	-0.19040	12.62385	0.13128	-0.12732	0	7264
KS11	0.00019	0.01715	0.00029	-0.22892	9.00933	0.11284	-0.12805	0	5536
KSE	0.00045	0.01465	0.00022	-0.32797	9.01387	0.12762	-0.13213	0	6391
MERV	0.00075	0.02301	0.00053	-1.80513	39.10840	0.16117	-0.47692	0	5692
MOEX10	0.00060	0.02032	0.00041	0.00791	27.54340	0.31256	-0.23483	0	4896
MXV	0.00049	0.01444	0.00021	0.01761	9.71147	0.12154	-0.14315	0	7039
PSELPS	0.00026	0.01409	0.00020	0.15793	12.38745	0.16178	-0.13089	0	7606
SS	0.00017	0.01546	0.00024	-0.33787	8.05874	0.09401	-0.09256	0	5618
TWII	0.00005	0.01378	0.00019	-0.20036	6.65229	0.08520	-0.09936	0	5530

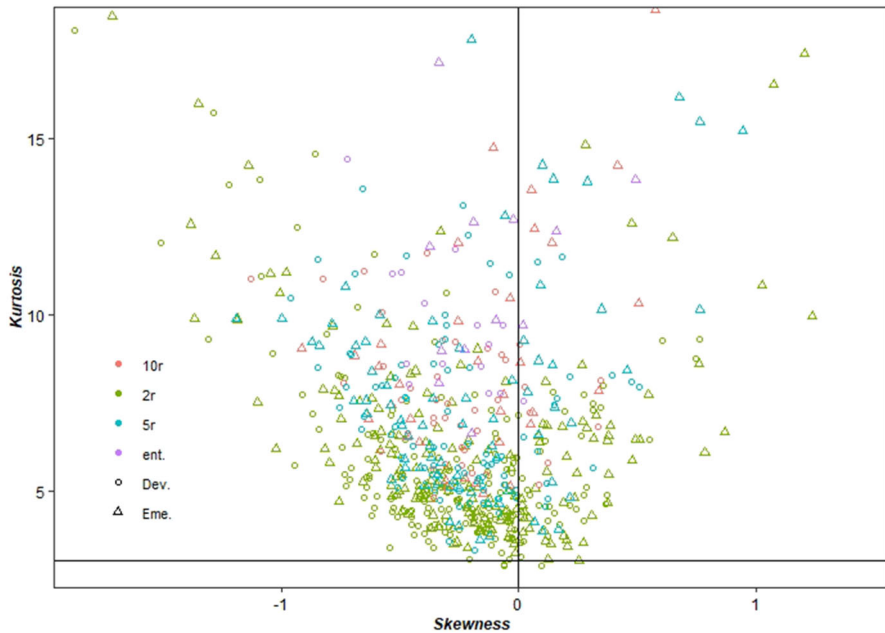


Fig. 2 Comparison of Skewness and Kurtosis of each series with different time interval for both developing (circle) and emerging markets (triangle)

where a significantly larger part of the individual series is characterized by a negative skewness. Such observations only confirm the findings of several current contributions dealing with the analysis of the empirical distribution of daily returns. Similar findings are reported in contributions such as Amaya et al. (2015), Lemperiere et al. (2017) some of them also point to skewness or asymmetry in the center of the distribution of daily returns Lemperiere et al. (2017) and Krämer (2021).

For the whole period, 83% of the markets are characterized by a negative value of the skewness, which indicates a longer left tail. Such a finding can also be observed in Fig. 2 where the largest cluster forms around the -0.3 level of skewness. From the point of view of individual subperiods, it is true that a negative degree of skewness can be observed in 75% of all subperiod series, it is also true that, 65% of the series with a positive skewness are developing markets.

For the whole data interval, the individual series are fairly symmetrical in only two cases, we can talk about moderately skewed data, namely in the case of GSPTSE and FTSEMIB. In one case, we can talk about highly skewed data, namely MERV. For each sub period, 73% of the series are fairly symmetrical, 21% of the series moderately skewed and 6% are highly skewed data. While 2.5 times more series of emerging markets are highly skewed compared to series of developed markets.

The findings, which have already been suggested by data on skewness and kurtosis, confirm the Jarque–Bera test of normality. In the case of the whole period, we reject the hypothesis of a normal distribution of data in all cases at the 1% level of significance. In the case of individual subperiods, we reject the hypothesis of a normal distribution

at the 1% level of significance in the case of two-year intervals in the 94% series, in the case of five-year intervals in the 99% series, in the case of ten-year intervals in the 100% series. This confirms the findings of a large number of other contributions dealing with the analysis of daily stock returns.

In order to examine whether other alternative distributions describe the leptokurtic characteristics of the empirical distribution of daily returns better compared to the normal distribution, in the following section we consider five alternative distributions. The selected distributions represent a mix of distributions that have been proposed in several contributions as possible alternative distributions of daily returns, where some alternatives are more flexible and provide parameters which allow modeling skewness and kurtosis, other are more restricted. Our aim is to examine the relative performance of selected alternatives (listed above) for modeling, that could better capture the characteristics of the empirical distributions compared to the normal distribution. As our goal is to apply the achieved results in the field of risk management, to the analysis of the goodness of fit of individual distributions, we also add a comparison of the behavior of individual models from the point of view of estimating risk measures as CVaR. All calculations contained in this article were performed in the RStudio environment.

4 Goodness of fit analysis

In the next section, we examine the suitability of individual distributions in comparison with the normal distribution. The parameters of individual distributions are estimated using the method of maximum likelihood estimation (ML), using Nelder-Mead or L-BFGS-B algorithm applied to the maximization of log likelihood function value. To compare the goodness of fit of individual distributions, we use quantified statistics namely Anderson–Darling statistics, Kolmogorov–Smirnov statistics, Akaike information criterion and Bayesian information criterion.

For the needs of comparison of the relative power of fit of individual distributions, we quantify the goodness of fit statistics, namely Anderson–Darling (AD) statistic and Kolmogorov–Smirnov (KS) statistic, as the selected distribution models differ in the number of parameters that define them, we also quantify information criteria that take into account the complexity of the model. Namely Akaike information criterion (AIC) and Bayesian information criterion (BIC).

In general, goodness of fit statistics measures the deviations of the fit between the distribution function (CDF) $\hat{F}$ and the empirical distribution function F_n . The general formula of Kolmogorov–Smirnov statistics is:

$$\sup \left| F_n(x) - \hat{F}(x) \right| \quad (9)$$

which represents the value of the greatest distance between the estimated and empirical distribution function. This statistic is more sensitive if the distributions differ near the center of the distribution.

On the other hand, Anderson–Darling statistics place more emphasis on adjustments in the tails of the distribution. The general formula of AD statistics is:

$$n \int_{-\infty}^{\infty} \frac{(F_n(x) - \hat{F}(x))^2}{\hat{F}(x)(1 - \hat{F}(x))} dx \quad (10)$$

For both of the above statistics, the smaller the value of the statistics, the better the power of fit to the empirical distribution. It is important to note that goodness of fit statistics do not penalize the number of distribution parameters in any way. Interpretation of these statistics when comparing models with different numbers of parameters can be complicated. As more parameters increase the flexibility of the model, on the other hand, it increases its complexity and overfitting may occur. To overcome this shortcoming, we also quantify AIC and BIC.

Table 5 contains the estimated values of individual parameters and the maximized value of the loglikelihood of individual selected distribution models for individual markets. The assumption of non-normal behavior of the data for whole period is also confirmed by the estimated values of the parameter p within the GED. Let us remind that the special cases of GED are Laplace distribution if $p = 1$ and normal distribution if $p = 2$. As we can see from estimated values of parameter p are close to 1, which indicates a greater similarity of data distribution with a Laplace distribution which is more pointed and has wider tails compared to the normal distribution. The parameters λ_3 and λ_4 of gld are negative in all series, which indicates the categorization of the distribution of daily returns into 1st class of shapes. In order for the k -th moment of gld to exist, it must hold that $\min(\lambda_3, \lambda_4) > -1/k$. The estimated values show that in some series fourth moment not exists and in some never third moment. Estimates of parameters of individual models for subseries due to the need to represent a large amount of data are not presented in this paper, but are available from the authors.

Tables 6 and 7 contains quantified AD, KS statistics and AIC, BIC information criteria for individual markets and models for the whole period divided into developing and developed markets. The highlighted values represent the values of the best models from the perspective of individual statistics. In the case of AD, the INF value in some cases of the normal distribution describe an incomparably worse quality of fit compared to other models. The unsuitability of the normal distribution as a model of the distribution of daily returns for the whole interval is also confirmed by the quantified statistics. The values of the individual statistics of the normal distribution are incomparably worse than in the case of the rest of the tested distributions.

Comparison of AD statistics from the point of view of the whole interval for individual markets listed in Tables 6 and 7 point to, that comparable relative power of fit for most developed markets is provided by, *sgt_1*, *sgt_2* and *gld*. The AD statistics of such models are similar and it is difficult to identify whether the differences between them are significant. The above also applies to AIC and BIC, for comparison of individual models we will consider the rule described in more detail in the paper (Burnham and Anderson 2004) that we will consider significant change in *AIC* compared to *AIC_{min}* of the best model for which it applies that $\Delta AIC > 10$. If the model has $\Delta AIC_i > 10$

Table 5 Estimated parameters of the models and calculated natural logarithm of likelihood function

	GED				Gld				Laplace			
	μ	σ_p	p	lnL	λ_1	λ_2	λ_3	λ_4	lnL	μ	σ	lnL
AEX	6.9E-04	0.0089	1.000	21,002	7.2E-04	217.99	-0.26	-0.20	21,056	6.9E-04	0.0089	21,002
ATX	6.3E-04	0.0092	1.027	20,178	8.6E-04	197.89	-0.22	-0.13	20,262	6.4E-04	0.0091	20,177
AXJO	4.7E-04	0.0070	1.115	22,721	5.9E-04	247.49	-0.16	-0.09	22,769	4.8E-04	0.0067	22,708
BFX	3.4E-04	0.0077	1.000	23,080	5.1E-04	241.76	-0.23	-0.18	23,121	3.4E-04	0.0077	23,080
BSEN	8.1E-04	0.0104	1.022	15,936	8.0E-04	173.64	-0.20	-0.16	15,965	8.4E-04	0.0103	15,936
BVSP	1.3E-03	0.0153	1.014	16,431	1.2E-03	118.43	-0.18	-0.18	16,492	1.2E-03	0.0152	16,430
CROBEX	2.7E-04	0.0070	1.000	16,035	3.0E-04	335.08	-0.35	-0.32	16,182	1.9E-04	0.0070	16,035
EGX30	6.4E-04	0.0117	1.041	15,408	8.3E-04	153.11	-0.19	-0.15	15,448	6.4E-04	0.0116	15,407
FCHI	3.7E-04	0.0100	1.090	22,333	5.1E-04	176.85	-0.17	-0.12	22,373	3.7E-04	0.0096	22,324
FTSE	3.2E-04	0.0078	1.058	24,249	3.8E-04	230.37	-0.18	-0.14	24,304	3.3E-04	0.0076	24,245
FTSEMIB	4.5E-04	0.0109	1.007	16,523	5.6E-04	164.95	-0.21	-0.14	16,537	4.5E-04	0.0108	16,523
GDAXI	7.8E-04	0.0098	1.020	22,249	7.3E-04	182.21	-0.20	-0.15	22,263	7.9E-04	0.0098	22,248
GSPC	5.4E-04	0.0074	1.000	24,307	6.6E-04	267.35	-0.27	-0.21	24,337	5.4E-04	0.0074	24,307
GSPTSE	5.4E-04	0.0064	1.000	25,621	6.8E-04	306.50	-0.28	-0.19	25,711	5.4E-04	0.0064	25,621
HSI	6.0E-04	0.0106	1.000	21,145	6.2E-04	177.42	-0.23	-0.19	21,175	5.8E-04	0.0106	21,145
IPSA	2.6E-04	0.0078	1.018	21,323	3.5E-04	229.40	-0.18	-0.17	21,395	2.6E-04	0.0078	21,323
JKSE	6.1E-04	0.0093	1.000	21,683	6.1E-04	224.30	-0.29	-0.25	21,764	5.9E-04	0.0093	21,683
KS11	6.3E-04	0.0113	1.000	15,455	8.2E-04	201.17	-0.36	-0.30	15,481	6.3E-04	0.0113	15,455
KSE	8.1E-04	0.0100	1.000	18,698	9.9E-04	194.57	-0.27	-0.20	18,713	8.1E-04	0.0100	18,698
MERV	1.2E-03	0.0155	1.000	14,073	1.4E-03	119.26	-0.22	-0.17	14,110	1.2E-03	0.0155	14,073
MOEX10	8.4E-04	0.0132	1.000	12,907	1.1E-03	148.62	-0.25	-0.21	12,987	8.5E-04	0.0132	12,907

Table 5 (continued)

GED	Gld				Laplace							
	μ	σ_p	p	lnL	λ_1	λ_2	λ_3	λ_4	lnL	μ	σ	lnL
MXX	5.3E-04	0.0100	1.003	20,511	5.7E-04	184.24	-0.20	-0.19	20,554	5.3E-04	0.0100	20,511
N225	1.6E-04	0.0111	1.094	20,989	2.6E-04	156.72	-0.15	-0.11	21,014	1.7E-04	0.0107	20,980
NZX50	7.0E-04	0.0052	1.079	18,284	7.3E-04	335.33	-0.18	-0.10	18,339	7.3E-04	0.0051	18,279
OMXC20	7.0E-04	0.0093	1.089	14,859	7.1E-04	191.81	-0.18	-0.12	14,894	7.5E-04	0.0089	14,853
OMXS30	5.8E-04	0.0106	1.087	22,345	5.4E-04	168.76	-0.17	-0.14	22,388	6.0E-04	0.0102	22,337
PSELPS	3.9E-10	0.0096	1.000	22,422	3.5E-04	188.55	-0.20	-0.18	22,443	1.0E-08	0.0096	22,422
SS	3.1E-04	0.0102	1.000	16,126	5.4E-04	193.84	-0.28	-0.24	16,105	2.7E-04	0.0104	16,127
TAI25.TA	1.8E-04	0.0095	1.000	16,417	8.0E-04	192.42	-0.22	-0.17	16,413	1.8E-04	0.0095	16,417
TWII	3.7E-04	0.0096	1.000	16,347	4.7E-04	200.00	-0.25	-0.20	16,337	3.6E-04	0.0096	16,347
sgt_1				sgt_1								
μ	σ	λ	lnL	q	lnL	μ	σ	λ	p	q	lnL	
AEX	1.8E-04	0.0146	0.069	1.44	21,052	2.0E-04	0.0135	-0.067	1.56	2.48	21,060	
ATX	2.7E-04	0.0136	-0.084	1.68	20,262	2.7E-04	0.0136	-0.084	2.01	1.66	20,262	
AXJO	2.5E-04	0.0095	-0.070	2.05	22,769	2.4E-04	0.0094	-0.068	1.81	2.61	22,771	
BFX	1.9E-04	0.0121	-0.049	1.54	23,112	1.8E-04	0.0114	-0.042	1.55	2.81	23,121	
BSEN	4.0E-04	0.0156	-0.048	1.67	15,960	4.0E-04	0.0149	-0.047	1.57	2.98	15,967	
BVSP	1.2E-03	0.0228	-0.004	1.67	16,492	1.2E-03	0.0225	-0.003	1.87	1.91	16,493	
CROBEX	2.9E-04	0.0175	-0.001	1.11	16,178	2.8E-04	0.0139	0.001	1.75	1.41	16,180	
EGX30	5.7E-04	0.0172	-0.026	1.72	15,443	5.4E-04	0.0167	-0.025	1.70	2.51	15,446	
FCHI	1.6E-04	0.0139	-0.049	1.90	22,370	1.6E-04	0.0136	-0.045	1.70	2.84	22,374	

Table 5 (continued)

	sgt_1				sgt_1						
	μ	σ	λ	q	lnL	μ	σ	λ	p	q	lnL
FTSE	1.5E-04	0.0111	-0.040	1.78	24,303	1.5E-04	0.011	-0.040	1.81	2.22	24,304
FTSEMIB	-1.8E-05	0.0163	-0.063	1.69	16,529	-1.9E-05	0.0155	-0.057	1.38	4.64	16,547
GDAXI	2.3E-04	0.0147	-0.062	1.70	22,255	2.5E-04	0.0139	-0.063	1.36	4.85	22,278
GSPC	3.0E-04	0.0124	-0.052	1.40	24,324	3.1E-04	0.0110	-0.040	1.33	3.77	24,351
GSPTSE	2.4E-04	0.0104	-0.079	1.44	25,706	2.4E-04	0.0098	-0.073	1.69	2.04	25,710
HSI	3.4E-04	0.0166	-0.030	1.54	21,167	3.3E-04	0.0155	-0.028	1.43	3.44	21,185
IPSA	3.1E-04	0.0115	-0.007	1.73	21,398	3.1E-04	0.0114	-0.007	1.91	1.92	21,398
JKSE	3.1E-04	0.0170	-0.034	1.28	21,756	3.2E-04	0.0148	-0.032	1.54	2.19	21,765
KS11	5.2E-05	0.0281	-0.062	1.12	15,467	2.1E-04	0.0175	-0.038	1.09	5.32	15,506
KSE	5.6E-04	0.0168	-0.046	1.41	18,697	5.1E-04	0.0147	-0.038	1.28	4.61	18,721
MERV	8.3E-04	0.0241	-0.045	1.57	14,103	8.4E-04	0.0227	-0.042	1.53	2.99	14,112
MOEX10	7.9E-04	0.0212	-0.024	1.45	12,985	7.8E-04	0.0209	-0.023	1.93	1.56	12,986
OMXX	4.4E-04	0.0154	-0.016	1.59	20,549	4.6E-04	0.0147	-0.014	1.65	2.44	20,554
N225	-5.1E-05	0.0152	-0.039	1.99	21,011	-5.4E-05	0.0149	-0.035	1.58	3.71	21,020
NZX50	4.0E-04	0.0072	-0.087	1.94	18,343	4.1E-04	0.0072	-0.086	1.91	2.15	18,343
OMXC20	3.1E-04	0.0130	-0.058	1.86	14,892	3.0E-04	0.0127	-0.060	1.72	2.67	14,894
OMXS30	2.1E-04	0.0149	-0.044	1.82	22,385	2.3E-04	0.0146	-0.043	1.72	2.61	22,389
PSELPS	2.8E-04	0.0147	-0.009	1.62	22,438	2.8E-04	0.0140	0.003	1.46	3.43	22,450
SSS	2.9E-04	0.0187	-0.022	1.32	16,089	2.6E-04	0.0155	0.000	1.00	11.61	16,148
TAI25.TA	5.6E-04	0.0148	-0.027	1.58	16,402	4.9E-04	0.0137	0.016	1.14	8.92	16,424
TWII	3.7E-05	0.0158	-0.050	1.44	16,324	5.9E-05	0.0138	-0.037	1.15	8.31	16,355

Table 6 Evaluated goodness of fit statistics and information criterions for each model of developed markets

		AD	KS	AIC	BIC	AD	KS	AIC	BIC	
AEX	GED	4.9437	0.0217	-41,998.1	-41,977.6	3.0665	0.0142	-48,607.9	-48,587.1	GSPC
	gld	0.5030	0.0075	-42,109.8	-42,103	1.5860	0.0124	-48,671.4	-48,664.4	gld
	laplace	4.9443	0.0216	-42,000.1	-41,986.4	3.0590	0.0142	-48,609.9	-48,596	laplace
	norm	109.8089	0.0796	-40,406.3	-40,392.6	INF	0.0886	-46,735.1	-46,721.3	norm
	sgt_1	0.5734	0.0099	-42,095.7	-42,068.3	2.4919	0.0160	-48,639.9	-48,612.2	sgt_1
	sgt_2	0.1397	0.0044	-42,110.2	-42,076	0.7214	0.0101	-48,692.2	-48,657.5	sgt_2
ATX	GED	6.3981	0.0209	-40,349.6	-40,329.2	5.9065	0.0209	-51,236.3	-51,215.5	GED
	gld	0.4994	0.0088	-40,522.6	-40,515.8	0.4498	0.0084	-51,419.6	-51,412.7	gld
	laplace	6.8417	0.0210	-40,349.9	-40,336.3	5.9550	0.0211	-51,238.3	-51,224.4	laplace
	norm	INF	0.0762	-39,079.5	-39,065.8	INF	0.0886	-49,174.5	-49,160.6	norm
	sgt_1	0.5666	0.0084	-40,515.6	-40,488.3	0.8480	0.0109	-51,403.1	-51,375.4	sgt_1
	sgt_2	0.5598	0.0082	-40,513.6	-40,479.5	0.6137	0.0098	-51,409.3	-51,374.7	sgt_2
AXJO	GED	3.4320	0.0175	-45,435.2	-45,414.7	1.6238	0.0122	-41,972.2	-41,951.5	GED
	gld	0.3568	0.0062	-45,536	-45,529.2	0.6786	0.0084	-42,026.8	-42,019.9	gld
	laplace	6.1759	0.0189	-45,411	-45,397.3	3.5643	0.0153	-41,955.3	-41,941.5	laplace
	norm	52.7614	0.0572	-44,548.6	-44,535	INF	0.0588	-40,993.9	-40,980.1	norm
	sgt_1	0.4640	0.0078	-45,531	-45,503.7	1.1476	0.0109	-42,013.7	-41,986.1	sgt_1
	sgt_2	0.3187	0.0065	-45,531.8	-45,497.6	0.3174	0.0060	-42,029.8	-41,995.3	sgt_2
BFX	GED	3.3329	0.0165	-46,154.4	-46,133.7	4.6627	0.0186	-36,562.9	-36,543.3	GED
	gld	0.6357	0.0076	-46,239.3	-46,232.4	0.3630	0.0076	-36,676.6	-36,670.1	gld
	laplace	3.3322	0.0166	-46,156.4	-46,142.6	6.5129	0.0243	-36,554.9	-36,541.9	laplace

Table 6 (continued)

	AD	KS	AIC	BIC	AD	KS	AIC	BIC	
FCHI	norm	0.0803	– 44,700	– 44,686.2	INF	0.0609	– 35,749.8	– 35,736.8	norm
	sgt_1	0.0098	– 46,215.3	– 46,187.7	0.1459	0.0072	– 36,677.4	– 36,651.3	sgt_1
	sgt_2	0.0096	– 46,231.3	– 46,196.8	0.1163	0.0058	– 36,675.9	– 36,643.2	sgt_2
	GED	0.0141	– 44,660.4	– 44,639.6	2.7886	0.0191	– 29,711.2	– 29,691.7	GED
	gld	0.0060	– 44,743	– 44,736.1	0.1929	0.0069	– 29,786.3	– 29,779.8	gld
	laplace	0.0199	– 44,644.3	– 44,630.4	4.3380	0.0270	– 29,701.3	– 29,688.3	laplace
FTSE	norm	0.0613	– 43,634.1	– 43,620.2	44.4615	0.0624	– 29,016.1	– 29,003.1	norm
	sgt_1	0.0082	– 44,731.8	– 44,704	0.3259	0.0092	– 29,775.9	– 29,749.9	sgt_1
	sgt_2	0.0061	– 44,737.5	– 44,702.9	0.1903	0.0092	– 29,778.4	– 29,745.9	sgt_2
	GED	0.0163	– 48,491.6	– 48,470.8	3.6991	0.0170	– 44,684.6	– 44,663.7	GED
	gld	0.0071	– 48,606.5	– 48,599.6	0.3616	0.0058	– 44,773.7	– 44,766.8	gld
	laplace	0.0193	– 48,485.3	– 48,471.4	5.5795	0.0221	– 44,669.1	– 44,655.2	laplace
FTSEMIB	norm	0.0621	– 47,276.4	– 47,262.5	68.7324	0.0567	– 43,628.1	– 43,614.2	norm
	sgt_1	0.0071	– 48,598.1	– 48,570.3	0.2554	0.0055	– 44,762.9	– 44,735	sgt_1
	sgt_2	0.0059	– 48,598.8	– 48,564.1	0.1542	0.0041	– 44,767.3	– 44,732.5	sgt_2
	GED	0.0160	– 33,039.8	– 33,019.8	6.7626	0.0332	– 32,828.7	– 32,808.8	GED
	gld	0.0109	– 33,071.5	– 33,064.9	1.9267	0.0230	– 32,823.9	– 32,817.3	gld
	laplace	0.0160	– 33,041.7	– 33,028.3	6.7709	0.0333	– 32,830.7	– 32,817.5	laplace
	norm	0.0712	– 32,005.6	– 31,992.2	67.9633	0.0750	– 31,789.3	– 31,776.1	norm
	sgt_1	0.0135	– 33,050.4	– 33,023.8	2.4800	0.0237	– 32,796.3	– 32,769.8	sgt_1
	sgt_2	0.0066	– 33,083.9	– 33,050.5	3.3277	0.0255	– 32,838.6	– 32,805.5	sgt_2
									TA125.TA

Table 6 (continued)

		AD	KS	AIC	BIC	AD	KS	AIC	BIC
GDAXI	GED	4.0367	0.0144	- 44,491.5	- 44,470.7				
	gld	1.3083	0.0103	- 44,523.2	- 44,516.2				
	laplace	4.3882	0.0152	- 44,492.6	- 44,478.7				
	norm	78.8847	0.0683	- 43,273.3	- 43,259.4				
	sgt_1	1.7875	0.0121	- 44,503	- 44,475.3				
	sgt_2	0.1758	0.0048	- 44,546.4	- 44,511.8				

Table 7 Evaluated goodness of fit statistics and information criterions for each model of emerging markets

		AD	KS	AIC	BIC	AD	KS	AIC	BIC	
BSESN	GED	2.5574	0.0148	-31,866.4	-31,846.5	2.9277	0.0151	-37,390.1	-37,369.8	KSE
	gld	0.2419	0.0072	-31,927	-31,920.4	1.2772	0.0125	-37,423.2	-37,416.4	GED
	laplace	2.9460	0.0150	-31,867.5	-31,854.3	2.9199	0.0151	-37,392.1	-37,378.6	gld
	norm	INF	0.0657	-30,905.3	-30,892.1	INF	0.0869	-36,022.5	-36,008.9	laplace
	sgt_1	0.4270	0.0090	-31,912.8	-31,886.3	2.1277	0.0152	-37,385	-37,357.9	norm
	sgt_2	0.1054	0.0046	-31,923	-31,889.9	1.1917	0.0120	-37,432.1	-37,398.2	sgt_1
BVSP	GED	3.4404	0.0157	-32,855.3	-32,834.9	2.0019	0.0166	-28,140.7	-28,120.8	sgt_2
	gld	0.4235	0.0071	-32,982.5	-32,975.7	0.3544	0.0090	-28,218.5	-28,211.9	GED
	laplace	3.6750	0.0155	-32,856.8	-32,843.3	1.9993	0.0166	-28,142.7	-28,129.4	gld
	norm	INF	0.0676	-31,521.8	-31,508.2	83.0788	0.0781	-26,785	-26,771.7	laplace
	sgt_1	0.3389	0.0065	-32,976.7	-32,949.5	0.6658	0.0116	-28,198.9	-28,172.4	norm
	sgt_2	0.3230	0.0059	-32,975.7	-32,941.7	0.1688	0.0062	-28,214.2	-28,180.9	sgt_1
CROBEX	GED	8.4925	0.0284	-32,063.9	-32,044.4	3.5576	0.0190	-25,807.2	-25,787.7	sgt_2
	gld	0.3949	0.0092	-32,361.4	-32,354.9	0.4814	0.0097	-25,971.2	-25,964.7	GED
	laplace	8.6429	0.0306	-32,066.4	-32,053.4	3.5582	0.0190	-25,809.2	-25,796.2	gld
	norm	INF	0.1228	-29,807	-29,794	INF	0.0951	-24,255.9	-24,243	laplace
	sgt_1	0.3946	0.0097	-32,348.9	-32,322.9	0.5245	0.0084	-25,962.9	-25,937	norm
										sgt_1

Table 7 (continued)

	AD	KS	AIC	BIC	AD	KS	AIC	BIC	
EGX30	sgt_2	0.3218	0.0080	-32,350.3	-32,317.8	0.5167	0.0088	-25,961.2	sgt_2
	GED	2.2567	0.0161	-30.811	-30,791.1	2.5884	0.0159	-41,016.6	GED
	gld	0.3627	0.0073	-30,894.4	-30,887.8	0.2164	0.0065	-41,106	gld
	laplace	2.7502	0.0188	-30,809.9	-30,796.6	2.6167	0.0160	-41,018.5	laplace
	norm	INF	0.0710	-29,837.9	-29,824.7	INF	0.0677	-39,678.5	norm
	sgt_1	0.6075	0.0092	-30,877.1	-30,850.6	0.3609	0.0083	-41,089.7	sgt_1
	sgt_2	0.4737	0.0075	-30,881.1	-30,848	0.1771	0.0054	-41,097.2	sgt_2
HSI	GED	1.7075	0.0113	-42,283.1	-42,262.4	5.4213	0.0263	-44,837.5	GED
	gld	1.0271	0.0108	-42,348.8	-42,341.9	1.0785	0.0186	-44,883.5	gld
	laplace	1.6663	0.0114	-42,285.2	-42,271.3	5.4255	0.0263	-44,839.5	laplace
	norm	INF	0.0770	-40,624.3	-40,610.5	INF	0.0675	-43,249	norm
	sgt_1	1.7521	0.0140	-42,325.4	-42,297.7	1.4168	0.0186	-44,867.4	sgt_1
	sgt_2	0.3702	0.0068	-42,359.8	-42,325.2	1.0440	0.0160	-44,889.5	sgt_2
								-44,854.8	PSEI,PS

Table 7 (continued)

		AD	KS	AIC	BIC	AD	KS	AIC	BIC	
IPSA	GED	3.5075	0.0204	-42,640.2	-42,619.7	2.0194	0.0259	-32,245	-32,225.1	GED SS
	gld	0.2768	0.0065	-42,788.3	-42,781.5	2.9035	0.0217	-32,209	-32,202.3	gld
	laplace	3.9016	0.0219	-42,641.4	-42,627.7	2.7180	0.0281	-32,249.1	-32,235.8	laplace
	norm	INF	0.0652	-41,197.1	-41,183.4	92.4366	0.0850	-30,904.1	-30,890.9	norm
	sgt_1	0.2227	0.0080	-42,787.4	-42,760.1	3.7569	0.0225	-32,170.6	-32,144.1	sgt_1
	sgt_2	0.1872	0.0069	-42,786	-42,751.9	1.4862	0.0272	-32,285.4	-32,252.3	sgt_2
JKSE	GED	4.7598	0.0192	-43,360.7	-43,340	1.9262	0.0148	-32,687.1	-32,667.2	GED TWII
	gld	0.2969	0.0073	-43,526.1	-43,519.2	1.3716	0.0138	-32,672.8	-32,666.2	gld
	laplace	4.9544	0.0196	-43,362.8	-43,349.1	1.9195	0.0147	-32,689.1	-32,675.8	laplace
	norm	INF	0.0918	-41,224.8	-41,211	73.2396	0.0784	-31,690.4	-31,677.2	norm
	sgt_1	0.6822	0.0104	-43,503.5	-43,475.9	1.9551	0.0173	-32,640.6	-32,614.2	sgt_1
	sgt_2	0.2367	0.0074	-43,520.4	-43,486	0.6325	0.0100	-32,700.3	-32,667.2	sgt_2
KS11	GED	7.1411	0.0267	-30,903.4	-30,883.5					
	gld	2.0284	0.0145	-30,961	-30,954.4					
	laplace	7.1382	0.0267	-30,905.4	-30,892.1					
	norm	123.8561	0.0988	-29,301.6	-29,288.3					
	sgt_1	2.5035	0.0183	-30,926.1	-30,899.6					
	sgt_2	0.8987	0.0103	-31,002.6	-30,969.5					

compared to AIC_{min} of the best model, it is unlikely that the i -th model would be a suitable description of the empirical data. We will apply such a rule also in the case of comparison of individual models using BIC.

In accordance with the above rule, we can mark sgt_2 and gld distributions as models with the best quality of adaptation. Which, in accordance with individual statistics, reach comparable values. In several cases, we may also mark sgt_1 as relevant. The worst values in individual statistics can be observed in the case of a normal distribution. Which we cannot identify as a relevant alternative in either case. In accordance with the AIC, we can mark gld as a relevant alternative for 73% of the series, sgt_2 for 90% of the series, sgt_1 for 23%, GED and Laplace distributions for 3% of the series. For BIC values, the absolute differences compared to the best distribution are clearer in favor of a single distribution. In the case of 93% of the series, we can mark gld distribution as relevant alternative, sgt_2 in case of the 17% of the series, GED and Laplace in case of 7% of the series as a relevant alternative. In all cases is unlikely, that norm a sgt_1 would be a suitable description.

From the point of view of all 639 subperiods, we can assess the relative stability of individual models when fitting data of different lengths and possible shapes of distributions. A comparison of the quantified AIC values shows that sgt_2 is a relevant alternative for 97% of the series, gld for 97% of the series, sgt_1 84% series, GED for 49%, Laplace for 30% and norm for 4% of the series. While at two-year intervals several models provide comparable quality of fit, on the other hand, at ten-year intervals only sgt_2 and gld offer comparable quality. Gld provides a better quality of fit for developed markets, on the other hand, for emerging markets sgt_2 provides better quality of fit. From the point of view of BIC, it is gld relevant alternative for 98.6% of the series, Laplace for 18%, GED for 11%, norm for 2%, sgt_2 for 1%, and sgt_1 for 0%. Regardless of the length of the intervals, it clearly that gld provides the best quality of fit according to BIC.

There are no significant differences in the results achieved, from the point of view of emerging and developed markets. However, in developed markets according to the AIC, gld provides a better fit for more series on the other hand, in emerging markets sgt_2 provides better fit for more series. According to the BIC, the best distribution is clearly gld , regardless of market segmentation. It is also true that distributions that do not contain a parameter capturing skewness such as Laplace, GED, norm achieve, on average, better values in developed markets compared to developing ones in. AIC and BIC values of individual subperiods are given in the Appendix.

5 CVaR estimation analysis

In this section, we will look at the comparison of individual models from the perspective of estimating the level of conditional value at risk (CVaR) in comparison with the historical CVaR calculated from empirical data.

VaR is a measure of risk that is generally defined as the maximum possible loss over a given period of time over a certain confidence interval (Yamai and Yoshida 2002). Let X be a random variable representing returns and $L(X)$ be a loss function,

then Value at Risk (VaR) can be mathematically defined as follows:

$$P(L(X) < VaR_\alpha) = 1 - \alpha \quad (11)$$

It follows from the definition, that VaR represents a value of loss where the possible loss will not exceed VaR with probability α . VaR can also be obtained by solving the problem:

$$VaR_\alpha = \inf\{l \in R : P(L(X) < l) \geq 1 - \alpha\}, \alpha \in (0, 1) \quad (12)$$

Value at risk (VaR) has become a standard tool for risk management in the financial sphere, mainly due to its conceptual and computational simplicity. However, many authors argue that VaR has several conceptual problems. The main critics of VaR include (Artzner et al. 1999; Basak and Shapiro 2001; Rootzén and Klüppelberg 1999). To mitigation the problems posed by VaR, is proposed an alternative measure of risk, called conditional value at risk (CVaR). CVaR is defined as the expected loss exceeding the VaR value. Based on this definition, CVaR only takes into account loss values that are higher than the VaR value. The CVaR value is defined as:

$$CVaR_\alpha(X) = E(L(X)|L(X) \geq VaR_\alpha) \quad (13)$$

where X denotes the random variable representing the return, $L(X)$ denotes the loss function of the random variable X and VaR_α is the value at risk described in the previous section. CVaR is a commonly used measure in the area of financial risk used to assess market risk and possibly in portfolio optimization e.g., in Ágoston (2012), Kara et al. (2019). $CVaR$ at the level α can be interpreted as the expected return in the case of $\alpha\%$ worst cases. Compared to the VaR measure, CVaR is more sensitive to the shape of the tail of the probability distribution and also provides better properties because it is a coherent and spectral risk measure. In addition, CVaR has properties as translational invariance, positive homogeneity, monotonicity, and subadditivity.

Quantified values of $CVaR_\alpha$ from empirical data for individual subperiods allow us to examine the dynamics of market risk from the perspective of developing and developed markets. Figure 3 captures the development of the average values of $CVaR_\alpha$ at the 1% level of significance within five-year intervals for developing and developed markets. From the Fig. 3 it is possible to observe convergence in the level of market risk between developing and developed markets.

As mentioned in this section, we will investigate the suitability of using distributions in determining the risk of market indices. When quantifying the risk measure $CVaR_\alpha$, we will consider significance levels of 5%, 1% and 0.5% which represent the risk associated with long positions (lower extreme of returns). In Figs. 4 and 5 we present the differences between the quantified values $CVaR_\alpha$ and the historical $CVaR_\alpha$ for the whole period at the significance level of 5%, 1%, 0.5%. Negative (positive) values of the difference between the estimated $CVaR_\alpha$ and the historical $CVaR_\alpha$ mean that the selected distribution overestimates (underestimates). We do not provide quantified values for individual subperiods, due to the need to present a large amount of data, but they are available from the authors.

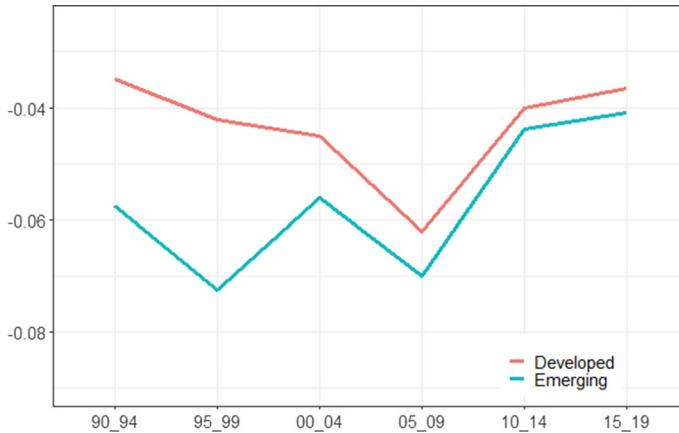


Fig. 3 The dynamics of market risk development on average in developing and developed markets

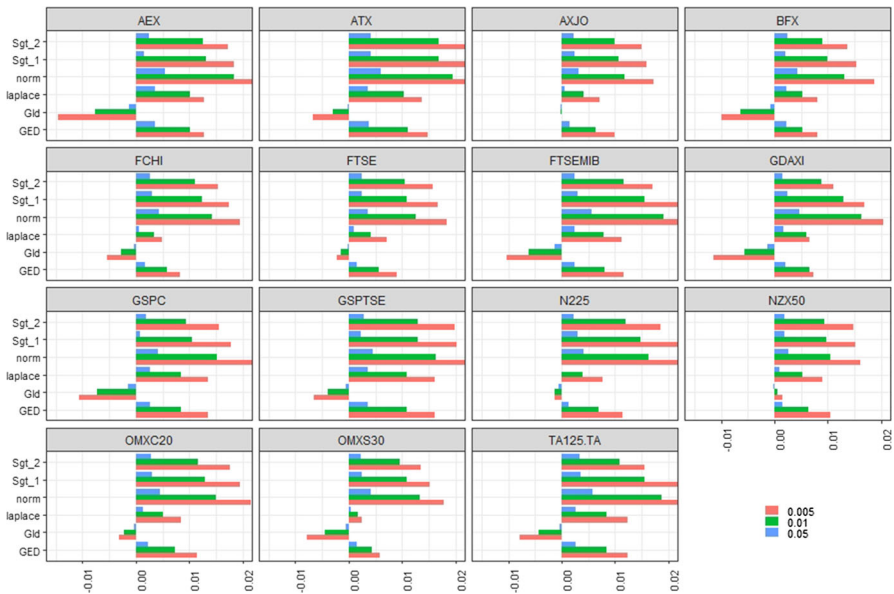


Fig. 4 Difference between estimated $CVaR$ and historical $CVaR$ for each model of daily returns of developing markets for whole period

From the comparison of quantified $CVaR_\alpha$ values shows that the gld distribution provides consistently the best estimates of $CVaR_\alpha$ values compared to other alternative distributions. This is true at all levels of significance. The lowest quality estimates it is provide at the 0.5% level of significance where for most cases it overestimates $CVaR_\alpha$ compared to $CVaR_\alpha$ quantified from empirical data. However, the greater the number of observations, the better estimates of $CVaR_\alpha$ gld provides. The normal

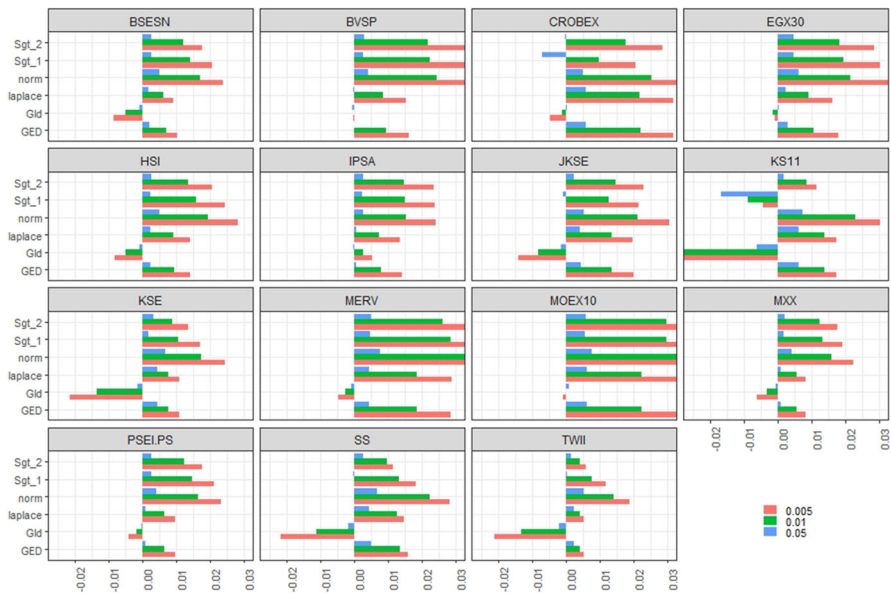


Fig. 5 Difference between estimated CVaR and historical CVaR for each model of daily returns of emerging markets for whole period

distribution provides stably the lowest quality of estimates. In terms of shorter periods, such as at two-year and five-year periods, the GED, Laplace, and sgt_2 distributions also provide comparable quality of the estimates.

From the point of view of developing and developed markets, it is true that the individual models provide better estimates in developed markets. This may be due to the fact that, we recall that CVaR is sensitive to the shape of the tails of distribution, that emerging markets are on average characterized by larger outliers than developed markets, which maybe partly seen from the comparison of min and max values in descriptive statistics. However, we also add that from the dynamics of the development of average values of min and max for developing and developed markets, it is also possible to observe convergence.

6 Conclusion

A number of authors have long been involved in the analysis of the distribution of financial asset returns. In the article, the authors deal with the analysis of the distribution of different markets (emerging and developed) from the 90 s of the twentieth century to the present. In addition to determining a more appropriate distribution of returns, we compare the suitability of their use in determining the value of CVaR risk measure.

The primary result of our contribution is the addition of further empirical evidence that daily returns are not normally distributed. The analyzed data show that the daily

returns of market indices of selected developing and developed markets are characterized by a slight skewness, mostly with negative skewness values, which means a longer left tail of empirical distribution and greater values of kurtosis indicating fatter tails of the distribution compared to the normal distribution.

In this paper, we comparing the relative performance of selected alternative distributions compared to the normal distribution in describing the leptokurtic behavior of daily returns. We perform the analysis both graphically using goodness of fit graphs and by comparing quantified goodness of fit statistics and information criteria. Finally, we also evaluate the applicability of our results to estimation of the selected measure of financial risk $CVaR_\alpha$.

The results of the comparison of selected distributions with the normal distribution can be summarize in the following conclusions. For daily return of market indices of selected countries, we reject the hypothesis of a normal distribution. The analyzed data are characterized by thicker tails compared to the normal distribution; in most cases we also observe negative values of skewness. All of the selected alternative distributions provide a more significant improvement in the quality of the fit compared to the normal distribution. The *sgt_2* and *gld* distributions are characterized by the best quality of fit from selected distributions. By AIC values, we can mark *sgt_2*, *gld* and *sgt_1* as a relevant alternative with comparable quality of fit. BIC values point more clearly to *gld*, where in accordance with the quantified values we can mark *gld* as a relevant alternative in 99% of all series. From a comparison of distributions for different long subperiods, *gld* provides a more stable quality of fit for different lengths of data compared to the rest of the distributions.

The results of the application of alternative distributions in estimating the degree of risk $CVaR_\alpha$, indicate a more significant improvement in quality compared to the normal distribution. Superior quality is provided by *gld*. The greater the number of observations is, the better the quality of the estimations compared to the historical $CVaR_\alpha$. The worst estimates are generated by *gld* at the 0.5% level of significance, where for most series it overestimates the values of $CVaR_\alpha$ compared to $CVaR_\alpha$ calculated from empirical data.

From the point of view of emerging and developed markets, we present the following findings. Emerging market data are, on average, more skewed than developed market data. However, in developing markets we can observe a significantly lower ratio of markets with a negative value of skewness than in the case of developed markets. Significant differences in terms of quality of fit of individual distributions between emerging and developed markets were not confirmed. From a comparison of the dynamics of market risk, we can observe convergence between emerging and developed markets. In line with the estimate of the financial risk level $CVaR_\alpha$, it follows that the individual selected distribution provides better estimates compared to the normal distribution regardless of the market type. On the other hand, estimates of individual models are, on average, better in developed markets than in developing markets.

As we have shown in our paper, *gld* describes the leptokurtic behavior of daily returns of market indices relatively well. The results of our work can bring additional attention to the expansion of the Generalized Lambda Distribution application in the field of Financial Modeling.

Acknowledgements This work was supported by the Grant Agency of Slovak Republic – VEGA grant no. 1/0339/20 “Hidden Markov Model Utilization in Financial Modeling”.

Appendix

See Figs. 6, 7, 8, 9, 10, 11, 12, and 13.

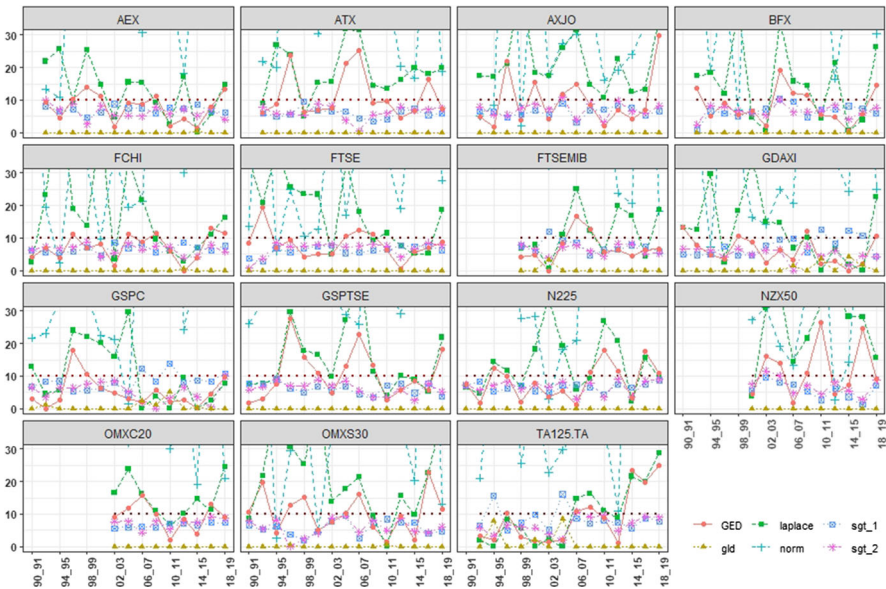


Fig. 6 Comparison of differences between AIC values of selected model and AIC of best model for individual two-year subperiods and developed markets. The interpretation of the graphs is as follows. For TA125.TA and subperiod 90_91 the model with the lowest AIC value is gld, for Laplace, sgt_1, sgt_2 and GED it is true that ΔAIC is less than 10, (on the graph highlighted by a black dotted line) which means we cannot unambiguously exclude the given models as unsuitable alternatives. The normal distribution provides a comparatively poorer quality of adaptation than other alternatives



Fig. 7 Comparison of differences between AIC values of selected model and AIC of best model for individual two-year subperiods and emerging markets



Fig. 8 Comparison of differences between BIC values of selected model and BIC of best model for individual two-year subperiods and developed markets

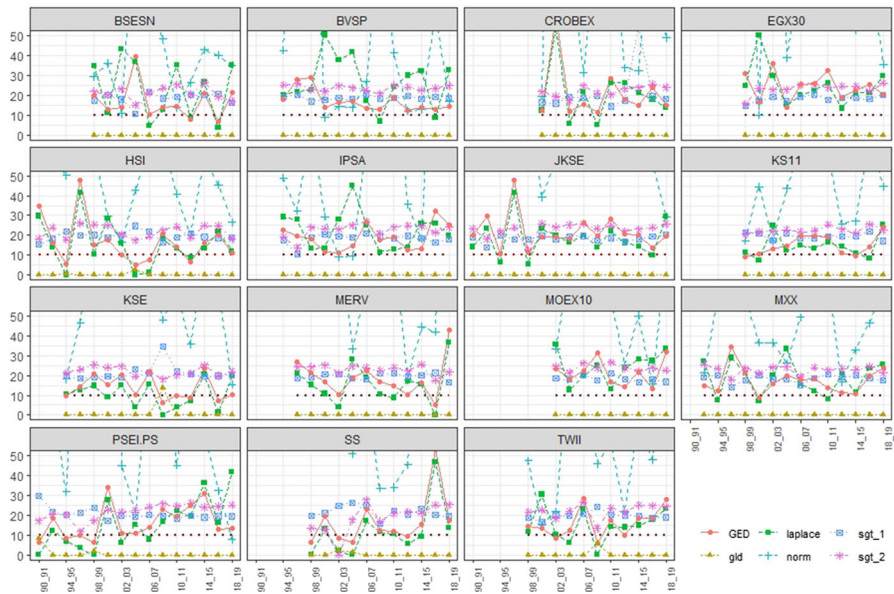


Fig. 9 Comparison of differences between BIC values of selected model and BIC of best model for individual two-year subperiods and emerging markets

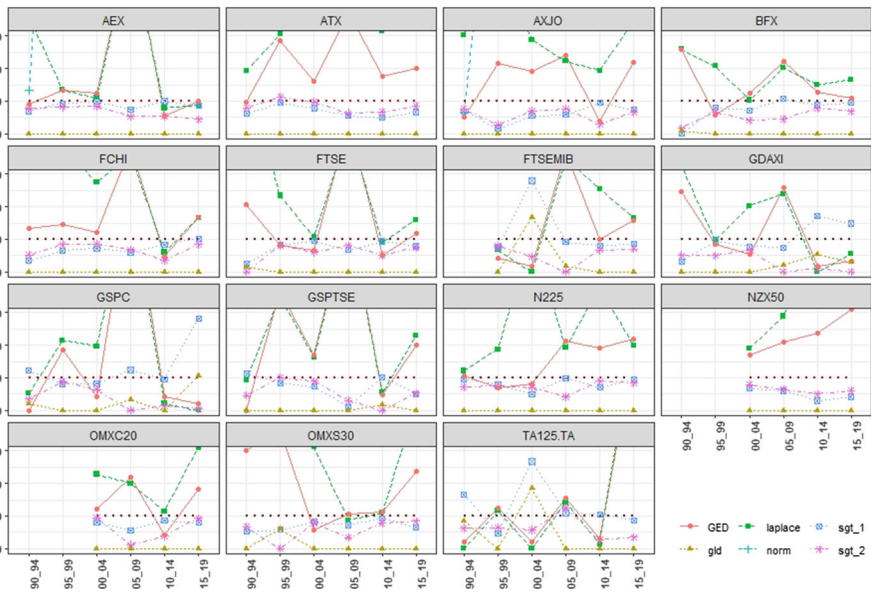


Fig. 10 Comparison of differences between AIC values of selected model and AIC of best model for individual five-year subperiods and developed markets

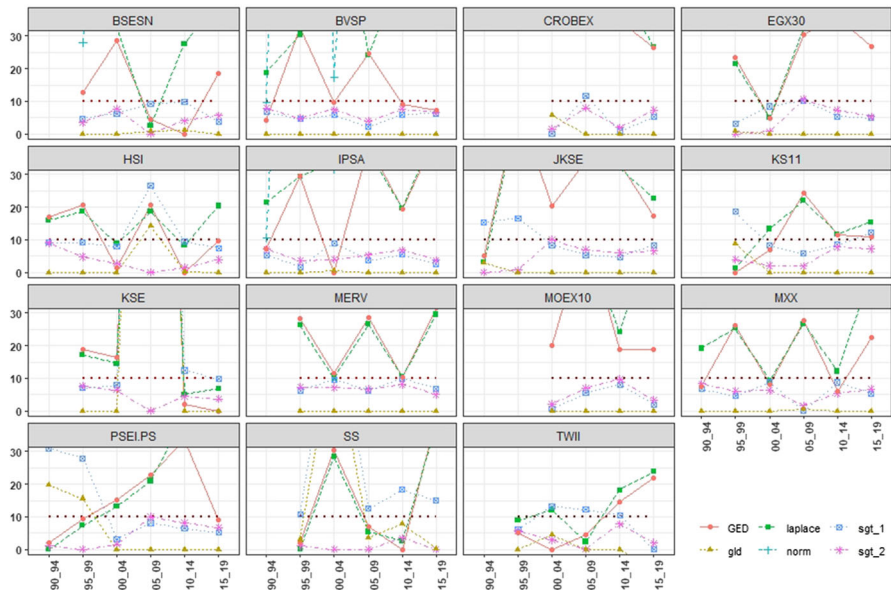


Fig. 11 Comparison of differences between AIC values of selected model and AIC of best model for individual five-year subperiods and emerging markets

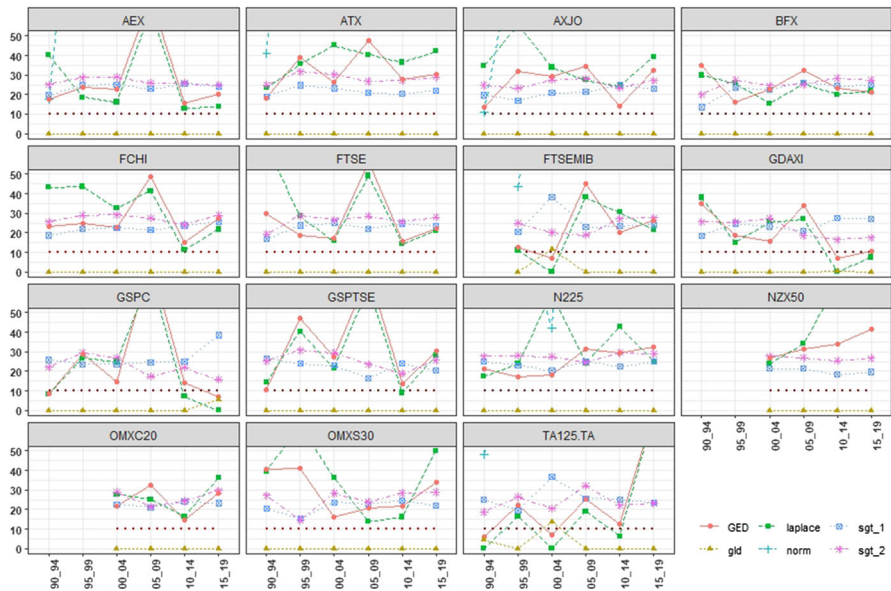


Fig. 12 Comparison of differences between BIC values of selected model and BIC of best model for individual five-year subperiods and developed markets

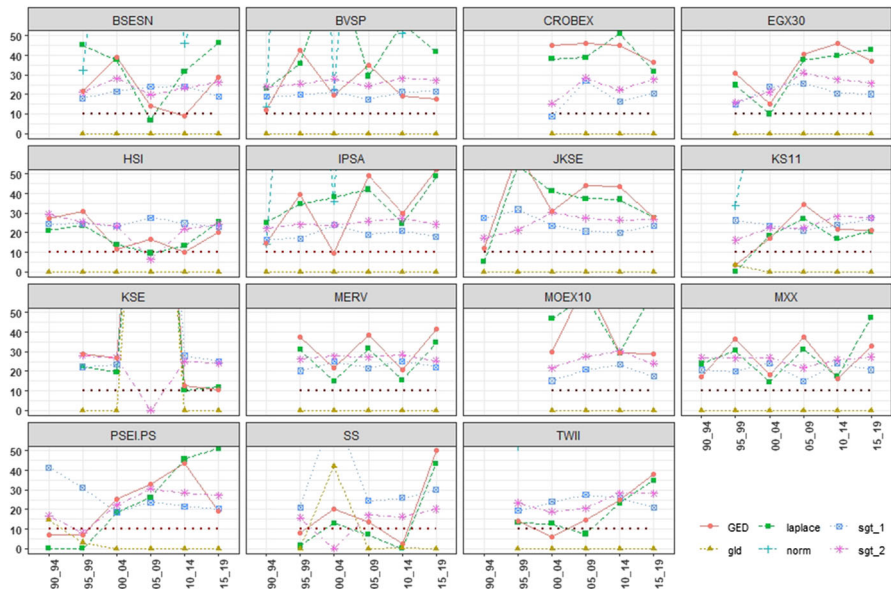


Fig. 13 Comparison of differences between BIC values of selected model and BIC of best model for individual five-year subperiods and emerging markets

References

- Ágoston K (2012) CVaR minimization by the SRA algorithm. *CEJOR* 20:623–632
- Amaya D, Christoffersen P, Jacobs K, Vasquez A (2015) Does realized skewness predict the cross-section of equity returns? *J Financ Econ* 118:135–167
- Artzner P, Delbaen F, Jean-Marc E, Heath DD (1999) Coherent measures of risk. *Math Financ* 9(3):203–228
- Basak S, Shapiro A (2001) Value-at-risk based risk management: optimal policies and asset prices. *Rev Financial Studies* 14(2):371–405
- Behr A, Pötter U (2009) Alternatives to the normal model of stock returns: Gaussian mixture, generalized logF and generalised hyperbolic models. *Ann Finance* 5(1):49–68
- Bollerslev T (1986) Generalized autoregressive conditional heteroscedasticity. *J Econom* 31(3):307–327
- Burnham PK, Anderson RD (2004) Multimodel inference: understanding AIC and BIC in model selection. *Sociol Methods Res* 33:261–304
- Corlu GC, Meterelliyoğlu M, Tiniç M (2016) Empirical distributions of daily equity index returns: a comparison. *Expert Syst Appl* 54:170–192
- Dean B (2013) Improved estimation and regression techniques with the generalized lambda distribution. PhD. The University of Newcastle, Australia, Faculty of Science and Information Technology
- Engle R (1982) autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica* 50(4):987–1007
- Fama E (1965) The behavior of stock market price. *J Bus* 38(1):34–105
- Freimer M, Kollia G, Mudholkar GS, Lin CT (1988) A study of the generalized Tukey lambda family. *Commun Stat-Theory Methods* 17(10):3547–3567
- Glawischning M, Seidl I (2013) Portfolio optimization with serially correlated, skewed and fat tailed index returns. *CEJOR* 21(1):153–176
- Gray JB, French DW (1990) Empirical comparisons of distributional models for stock index returns. *J Bus Financ Acc* 17(3):451–459
- Hansen BC, McDonald BJ, Newey KW (2010) Instrumental variables estimation with flexible distributions. *J Bus Econ Stat* 28(1):13–25

- Harris RD, Küçüközmen CC (2001) The empirical distribution of UK and US stock returns. *J Bus Financ Acc* 28(5):715–740
- Hull J, White A (1987) The pricing of options on assets with stochastic volatilities. *J Financ* 42(2):281–300
- Kara G, Özmen A, Weber GW (2019) Stability advances in robust portfolio optimization under parallel-leveled uncertainty. *Cent Eur J Oper Res* 27:241–261
- Krämer W (2021) Asymmetry in the distribution of daily stock returns. *Empir Econ* 60(1):1115–1125
- Lempérière Y, Deremble C, Nguyen TT, Seager P, Potters M, Bouchaud JP (2017) Risk premia: asymmetric tail risks and excess returns. *Quant Finance* 17(1):1–14
- Mandelbrot B (1963) The variation of certain speculative prices. *J Bus* 36(4):394–419
- Massing T (2019) What is the best Lévy model for stock indices? A comparative study with a view to time consistency. *Financ Mark Portf Manag* 33:277–344
- Peters E (1991) *Chaos and order in the capital markets. A new view of cycles, prices, and market volatility.* Wiley, New York
- Praetz P (1972) The distribution of share price changes. *J Bus* 45(1):49–55
- Rootzén H, Klüppelberg C (1999) A single number can't hedge against economic catastrophes. *Ambio* 28(6):550–555
- Theodossiou P (1998) financial data and the skewed generalized T distribution. *Manage Sci* 44(12):1650–1661
- Tukey JW (1960) The practical relationship between the common transformations of percentages or fractions and of amounts. Technical Report 36, Statistical Research Group, Princeton
- Yamai Y, Yoshida T (2002) On the validity of value-at-risk: comparative analyses with expected shortfall. *Monet Econ Stud* 20(1):57–85

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.